

Javier De Leon-Flores

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Education

University of Virginia, B.S. Mechanical & Aerospace Engineering (May 2028) | **M.E.** Mechanical & Aerospace Engineering (June 2028)

GPA: 3.597/4.00 | **Relevant Coursework:** Fluid Mechanics, Thermodynamics, Compressible Aerodynamics, Graduate Gas Dynamics

Related Experience

Project Lead – Gas-Fed Pulsed Plasma Thruster Capstone

August 2026 - Present

- Lead a 13-member team developing UVA's first electric thruster, coordinating thruster design, electronics, and system/testing subteams.
- Established project milestones, technical deliverables, and development schedules, providing cross-disciplinary technical guidance to enable independent subteam progress.
- Develop a 200-1500 V power processing unit (PPU) architecture integrating high-voltage conversion, energy storage, valve actuation, and microcontroller-based control.
- Investigate self-ignition and electric breakdown in a coaxial argon-fed thruster using reduced electric field (E/N) and CFD analysis to optimize gas distribution for azimuthally uniform electrical breakdown.
- Plan vacuum chamber testing and the development of custom Langmuir probes and a thrust stand to measure impulse bit, characterize plasma discharge, and determine specific impulse.

Undergraduate Research Assistant – Electric Propulsion and Magnetic Nozzle Physics (Chen Cui Lab)

June 2026 - Present

- Execute fully kinetic particle-in-cell (PIC) simulations on HPC systems and post-process simulation data in MATLAB to extract plasma properties and generate quantitative results.
- Analyze plasma properties to quantify the influence of high-energy tail electrons on thermodynamics and azimuthal current formation, contributing to a research paper in preparation for an AIAA conference.
- Developed a MATLAB sensitivity analysis of electron temperature and density contours, identifying the top 12% of electron energies as threshold separating tail and bulk populations.

MARS at UVA – NASA Lunabotics, Front Loader Subteam Lead

Sep 2024 – Present

- Achieved 1st overall at NASA Lunabotics 2026 and earned two additional competition awards among 38 international teams.
- Directed a 5-member subteam, integrating workstreams and leading documentation to meet design review deadlines.
- Reduced system mass by 20 kg through bucket geometry optimization and center-of-mass redistribution using SolidWorks analysis and hand calculations.
- Designed and executed a test setup to analyze the impact of vibration-induced fluidization on frictional resistance; results enabled reduction of deposit angle from 45° to 15°.
- Fabricated aluminum components via CNC milling, waterjet cutting, and metal rolling, contributing to final rover assembly.

Work Experience

Factory Direct Fastening – Mechanical Design Engineer (Electrical Box Lock)

May 2025 – Present

- Delivered proprietary design of a municipal electrical box lock, optimizing original concept for manufacturability, deployment, and theft deterrence.
- Designed a custom, assembly-integrated locking mechanism using a friction fit, eliminating the need for an off-the-shelf lock.
- Redesigned the locking assembly using a simplified finger-joint mechanism, reducing part count from 15 to 7.
- Optimized joint geometry through SolidWorks motion and tolerance analysis, eliminating visible gaps to create a seamless, solid-block appearance when locked.
- Produced complete engineering drawings and manufacturing specifications; collaborated with two manufacturers to refine geometry, reducing manufacturing cost by 10% and producing a fabricated prototype.

Skills

Technical: SolidWorks (CSWP), Fusion 360, Ansys, MATLAB, Python, C++, HPC

Manufacturing: CNC Milling, Waterjet, Sheet Metal, 3D Printing

Design and Analysis: FEA, GD&T, CFD, Tolerance Analysis, DFM/DFA